

KOHLER AI Bathroom Designer & Planner

From customer constraints
to a **physically feasible** bathroom design.

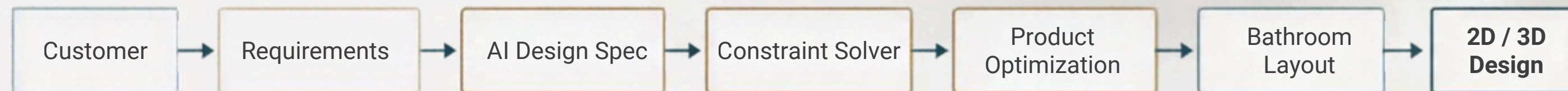
CUSTOMER INPUTS

Bathroom dimensions
Budget
Style preferences
Required fixtures
Optional floor plan/image

THE DIFFERENCE

Traditional systems
deliver product lists only.

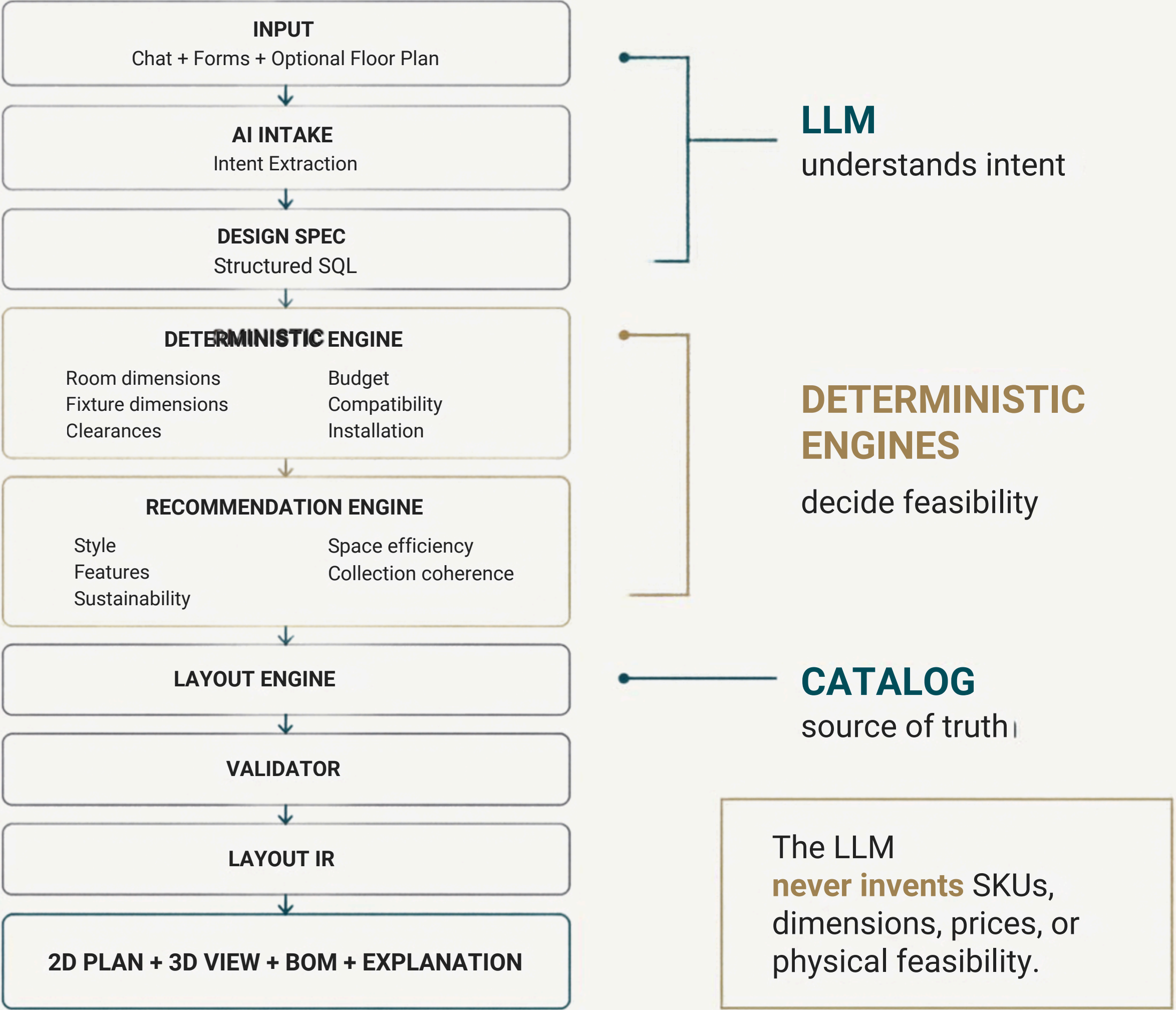
**We deliver a feasible
bathroom design.**



We don't just recommend products.
We compile customer requirements into a **feasible bathroom design**.

Architecture

From intent to installable
bathroom design



What Makes It Different

GENERIC AI ASSISTANT

Generates product suggestions

May hallucinate specifications

No physical validation

No guaranteed budget fit

No actual layout reasoning

vs.

OUR DESIGN COMPILER

Structured Design Spec

Catalog-grounded products

Hard constraint validation

Optimization of feasible configurations

Collision/clearance validation

One Layout IR drives 2D + 3D

Conversational redesign loop

AI RESPONSIBILITIES

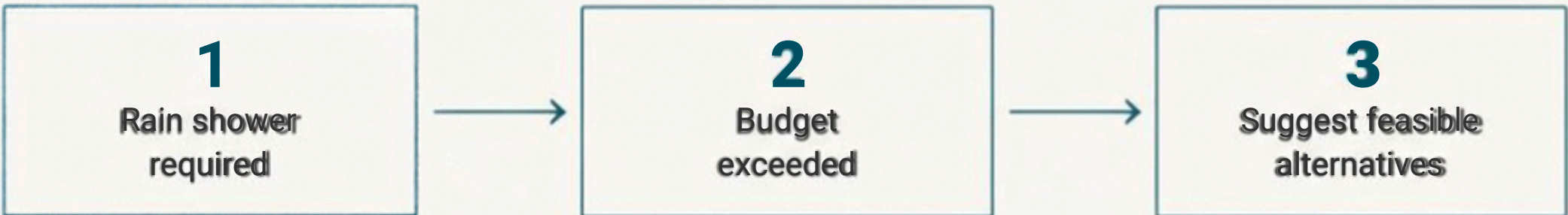
1 **Intent Extraction**
Natural language → Structured Design Spec

2 **Catalog RAG**
Retrieve grounded product information

3 **Design Copilot**
User says: "Make the shower bigger" and reduce the budget by 10%."
Intent → Constraint Update →
Re-solve → New Layout

Hard constraints
are **overridden by AI.**

REPAIR FLOW



Technical Execution & Real-World Impact

TECH STACK

Frontend: Next.js • React • TypeScript • Tailwind

Backend: FastAPI • Python

Data: PostgreSQL • pgvector

AI: LLM • Structured Outputs • RAG • Optional Vision

Visualization: SVG/Canvas 2D • Three.js / React Three Fiber

Infrastructure: Docker • GitHub • Vercel • Railroad/Render

ENGINEERING CAPABILITIES

Constraint-aware:
Physical feasibility before recommendation

Explainable:
Every recommendation has a reason

Sustainable:
Water efficiency incorporated into scoring

Scalable:
Catalog + solver + Layout IR architecture

KOHLER BUSINESS VALUE

Customer personalization

Feasible product bundles

Reduced decision friction

Water-efficient choices

Quote-ready BOM

Scalable digital design experience



AI generates the intent. **Engineering** guarantees the design.